Physics++

A Continuous-Limit Computational Framework for Information Fields

From Diffusion–Wave Equations to Computable Numerical Fields
—with Remarks on the Physical Foundations of Transformers

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摘要

This paper proposes a formal information-field framework grounded in the continuous-depth limit and variational principles. Starting from physical axioms concerning computational reducibility and dimensional algebra, we systematically reconstruct each component of the Transformer architecture within a physics-based analogy. By applying Euler integration, discrete layer stacking is mapped onto a continuous dynamical system. The principle of least action then yields a tensor diffusion–wave equation as the governing dynamics of the information field. These theoretical developments are accompanied by a computable numerical implementation provided by the Physics++ library. Numerical experiments conducted on a 512-point discrete field verify key physical properties—wave-packet formation, linear superposition, energy homogeneity ($E \propto A^2$), and resonant modulation—with additive errors below 1% and homogeneity errors below 0.5%. We do not claim that Transformers "are" physical fields; rather, we offer a reformulation of Transformer dynamics in the language of classical field theory, together with a numerically verifiable computational model.

Keywords: information field, Transformer, diffusion–wave equation, Euler–Lagrange equation, quaternion wave, computational physics, continuous-depth limit

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1 Introduction

Since its introduction by Vaswani et al. (2017), the Transformer architecture has fundamentally reshaped sequence modelling. Nevertheless, prevailing explanations remain largely engineering-oriented—terms such as "attention mechanism", "residual connection", and "layer normalisation" describe *how* computation proceeds, but seldom address *what* is being computed.

From a rigorous mathematical and physical standpoint, this work advances a central thesis: **Matrix computations within Transformers operate on an approximately gradient-like field, describable by a diffusion–wave equation, whose solution space constitutes an information field.**

Physics++ is a C++ numerical framework constructed upon first principles of physics. Its guiding philosophy is that any computable physical process admits a rigorous description via axiomatic foundations, continuous limits, and variational principles, and can be validated through executable numerical code.

Structure of this paper. Section 2 establishes the axiomatic basis (computational reducibility and dimensional algebra). Section 3 reconstructs each Transformer component within a physics-based formalism. Section 4 maps discrete layer stacking onto a continuous dynamical system via the continuous-depth limit. Section 5 derives the governing dynamics of the information field using Lagrangian variational principles. Section 6 provides a formal definition of the information field. Section 7 discusses relations to existing literature. Section 8 describes the numerical architecture of Physics++. Section 9 presents numerical experiments. Section 10 concludes with limitations and directions for future research.

2 Axiomatic Foundations

2.1 The Axiom of Computational Reducibility

Axiom 2.1 (Computability of Physical Computation)**.** *Any computable physical quantity must satisfy:*

1. ***Finite representation**: its domain and codomain are subsets representable by finite-precision floating-point numbers;*
2. ***Termination**: its evaluation terminates after finitely many steps;*
3. *ε**-consistency**: the discrepancy between the computed result and the continuum-theoretical value admits a computable upper bound ε.*

Mathematical justification. Conditions (1) and (2) follow directly from Turing computability theory: any computable function must halt within finitely many steps. Condition (3) originates from backward error analysis in numerical analysis (Wilkinson, 1963): a numerically stable algorithm computes

$$\mathrm{computed}(f(x)) = f(x + \delta x), \qquad \|\delta x\| \le \varepsilon \|x\|,$$

where ε denotes machine precision (for double precision, $2^{-53} \approx 1.1 \times 10^{-16}$). Thus, within machine tolerance, numerical computation is indistinguishable from exact mathematics.

Engineering realisation. Pre- and post-computation checks for NaN and Inf in `mathematics.h` implement condition (1): NaN (Not-a-Number) and Inf (Infinity) are non-computable states under IEEE 754 and violate the axiom of reducibility if propagated.

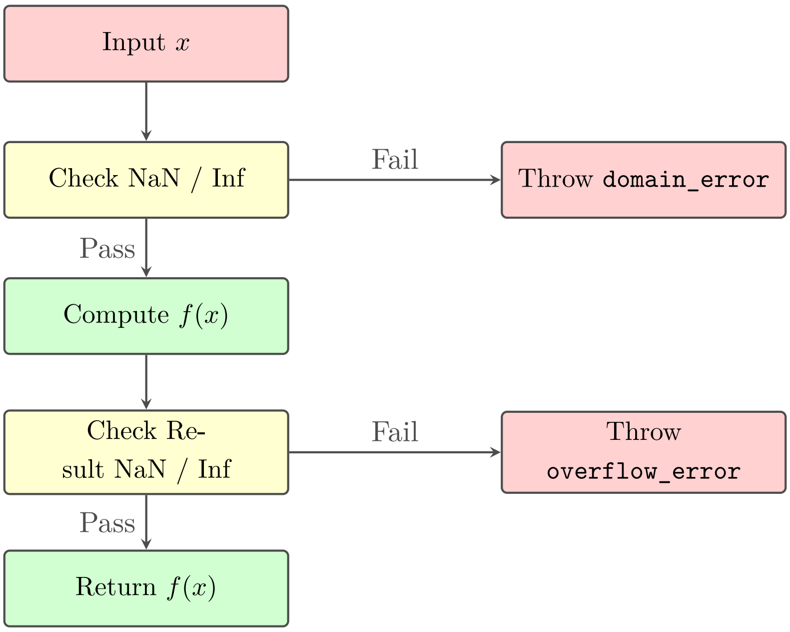

图 1: Safe numerical workflow: preconditions protect inputs (NaN/Inf detection); postconditions protect outputs (overflow detection).

2.2 The Axiom of Physical Algebra

Axiom 2.2 (Dimensional Algebra of Physical Quantities)**.** *Physical quantities form a graded commutative algebra*

$$\mathcal{A} = \bigoplus_{d \in \mathcal{D}} \mathcal{A}_d,$$

where $\mathcal{D}$ is a seven-dimensional vector space of dimensions corresponding to the SI base quantities: length L, mass M, time T, electric current I, thermodynamic temperature Θ, amount of substance N, and luminous intensity J. The algebraic operations satisfy:

- *Addition is defined only between quantities of identical dimension: $+ : \mathcal{A}_d \times \mathcal{A}_d \to \mathcal{A}_d$;*
- *Multiplication adds dimensions: $\times : \mathcal{A}_{d_1} \times \mathcal{A}_{d_2} \to \mathcal{A}_{d_1+d_2}$;*
- *Division subtracts dimensions: $\div : \mathcal{A}_{d_1} \times \mathcal{A}_{d_2} \to \mathcal{A}_{d_1-d_2}$.*

Mathematical justification. This is the algebraic formulation of the Buckingham π theorem (Buckingham, 1914): any physical law $f(q_1, \ldots, q_n) = 0$ is equivalent to a relation among dimensionless parameters $F(\pi_1, \ldots, \pi_{n-r}) = 0$, where r is the rank of the dimensional matrix. This guarantees that physical formulae are independent of the choice of units—one of the most fundamental symmetries in physics.

Engineering realisation. Each physical type in `quantization.h` (`Time`, `Voltage`, `Force`, etc.) implements dimensional algebra via C++ operator overloading. Compile-time type checking ensures that expressions such as `Time + Resistance` fail to compile—a direct encoding of Axiom 2.2 into the type system.

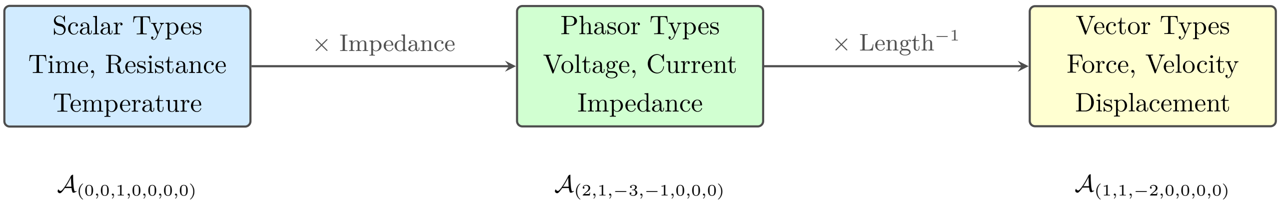

图 2: Dimensional algebra of physical types: each type belongs to a subspace of the dimensional space $\mathcal{D}$; multiplication across dimensions produces new physical types.

3 Physical Reconstruction of the Transformer Architecture

Before passing to the continuous limit, we examine each Transformer component in detail, exposing the underlying physical structure concealed beneath engineering terminology. The objective here is not to assert that Transformers "are" physical systems, but to establish a **precise formal analogy**—one that permits the application of mature mathematical tools from classical field theory to the analysis of information-processing dynamics.

3.1 Token Embeddings as Discrete Field Samples

Definition 3.1 (Token Embedding Field)**.** *Given a vocabulary $\mathcal{V}$ and an input sequence $(w_1, \ldots, w_n)$, the token embedding matrix $\mathbf{X} \in \mathbb{R}^{n\times d}$ defines a **discrete vector field**:*

$$\mathbf{X} : \{1, 2, \ldots, n\} \to \mathbb{R}^d, \qquad \mathbf{X}(i) = \mathbf{x}_i.$$

Here n corresponds to discrete spatial positions and d to the number of field components per position. In the continuous limit $n \to \infty$, $\mathbf{X}(i) \to \mathcal{I}(\mathbf{r})$, where $\mathbf{r} \in \mathcal{M}$ denotes coordinates on a semantic manifold.

Physical analogy. This resembles the **tight-binding model** in solid-state physics: each atomic site i hosts an orbital wavefunction ϕ_i, and the crystal wavefunction is a linear combination $\psi = \sum_i c_i\phi_i$. Token embeddings $\mathbf{x}_i$ correspond to atomic orbitals, while the attention matrix $\mathbf{A}$ corresponds to the hopping matrix t_{ij}.

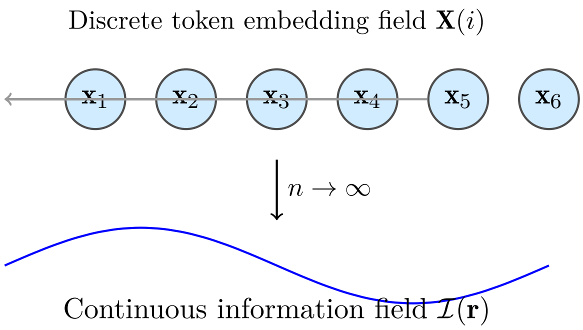

图 3: Transition from a discrete token embedding field to a continuous information field. Each token represents a sampling point; attention weights define coupling strengths between samples.

3.2 Inner-Product Similarity as a Hilbert-Space Metric

The core operation of attention scoring is

$$s_{ij} = \frac{\langle \mathbf{q}_i, \mathbf{k}_j \rangle}{\sqrt{d_k}},$$

where $\mathbf{q}_i = \mathbf{W}_Q \mathbf{x}_i$ and $\mathbf{k}_j = \mathbf{W}_K \mathbf{x}_j$.

Formal Analogy 3.1 (Inner Product). In quantum mechanics, the transition amplitude between two states $|\psi\rangle$ and $|\phi\rangle$ is proportional to their inner product $\langle \psi | \phi \rangle$ (Feynman, 1965). The attention score s_{ij} formally resembles the transition amplitude between tokens i and j within a projected Hilbert space $\mathcal{H}_{d_k}$. Division by $\sqrt{d_k}$ preserves variance—a standard practice rooted in random projection theory (Johnson–Lindenstrauss lemma, Dasgupta & Gupta, 2003).

Physical intuition. The attention mechanism defines a **query detector** at each token position: $\mathbf{q}_i$ encodes "what information the current token seeks", while $\mathbf{k}_j$ encodes "what information the candidate token offers". Their inner product measures the degree of match between them.

3.3 Softmax as a Partition Function

$$A_{ij} = \frac{\exp(s_{ij})}{\sum_k \exp(s_{ik})}.$$

Proposition 3.1 (Exact Softmax–Boltzmann Correspondence)**.** *Define energy $E_{ij} = -s_{ij}$ and temperature $kT = 1$. Then the Softmax function is exactly equivalent in form to the Boltzmann distribution of the canonical ensemble:*

$$A_{ij} = \frac{e^{-E_{ij}/kT}}{\sum_k e^{-E_{ik}/kT}}.$$

Mathematical justification. The fundamental postulate of statistical mechanics (Gibbs, 1902) states that the probability of a system occupying a state of energy E is proportional to $e^{-E/kT}$. Accordingly, attention weights admit interpretation as a **probability-mass distribution**: token i allocates its limited "attention mass" over tokens j, favouring low-energy (high-similarity) states and exponentially suppressing high-energy (low-similarity) ones.

Precision of this proposition. Proposition 3.1 follows directly from the definition of Softmax and involves no approximation. It constitutes the strongest physical correspondence in this paper—an exact formal equivalence rather than a heuristic analogy.

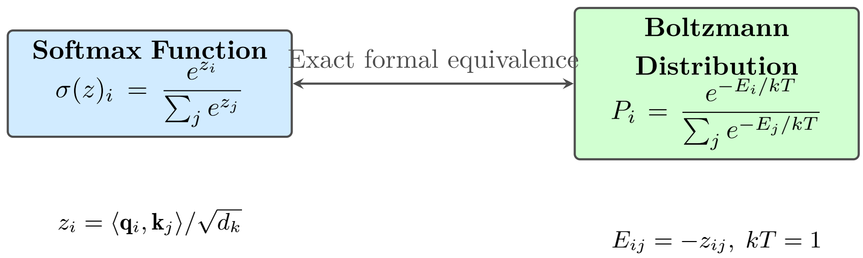

图 4: Exact formal equivalence between the Softmax function and the Boltzmann distribution. This is the strongest physical correspondence in this work.

3.4 Weighted Summation as a Statistical Expectation

$$\mathbf{y}_i = \sum_j A_{ij}\mathbf{v}_j.$$

Proposition 3.2 (Expectation Interpretation). *$\mathbf{y}_i$ is the **statistical expectation** of the value vectors $\mathbf{V}$ under the Boltzmann distribution $A_{i\cdot}$:*

$$\mathbf{y}_i = \mathbb{E}_{j\sim A_{i\cdot}}[\mathbf{v}_j].$$

Mathematical justification. This is the definition of the expectation of a discrete random variable. It reveals the essential nature of attention: rather than deterministically "selecting" the most similar token, the mechanism performs a probabilistic weighted average over all tokens' value vectors.

3.5 Residual Connections as Inertia

$$\mathbf{X}' = \mathbf{X} + \mathbf{F}(\mathbf{X}).$$

Proposition 3.3 (Residuals as Inertia). *The residual update may be rewritten as $\mathbf{X}' - \mathbf{X} = \mathbf{F}(\mathbf{X})$. The left-hand side is a **first-order discrete-time difference** $\Delta\mathbf{X}$, corresponding to the continuous-time velocity $\partial_t\mathbf{X}$. Residual connections ensure that $\mathbf{X}'$ does not deviate excessively from $\mathbf{X}$—the discrete analogue of **inertia** in physics: absent external force ($\mathbf{F} = \mathbf{0}$), a body maintains its present state ($\mathbf{X}' = \mathbf{X}$).*

Mathematical justification. This is the discrete form of Newton's first law. In the continuous limit, the residual connection becomes the Euler integration of the dynamical system $\partial_t\mathbf{X} = \mathbf{F}(\mathbf{X})$ —the central observation of Neural ODEs (Chen et al., 2018).

3.6 Layer Normalisation as Constraint Projection

$$\mathbf{X}' = \frac{\mathbf{X} - \mu(\mathbf{X})}{\sigma(\mathbf{X})} \odot \gamma + \beta.$$

Proposition 3.4 (Geometric Meaning of Layer Norm). *Layer normalisation projects each token embedding vector onto the **unit hypersphere** S^{d-1} (normalising to zero mean and unit variance), followed by an affine transformation. This is equivalent to imposing dynamics on a **constrained manifold**: the state of the information field is confined to a compact subset, preventing divergence of embedding norms and ensuring numerical stability.*

Physical analogy. In classical field theory, the **nonlinear σ model** (Gell-Mann & Lévy, 1960) constrains field values to a target manifold (e.g. S^2). Layer normalisation similarly restricts token embeddings to a compact subset of $\mathbb{R}^d$, providing a numerical guarantee of boundedness for information-field dynamics.

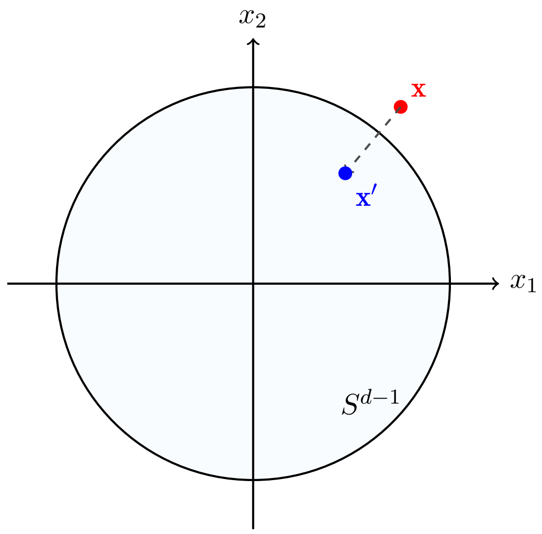

图 5: Geometric interpretation of layer normalisation: embedding vectors are projected onto the unit hypersphere, bounding the norm of the field.

3.7 Positional Encoding as Initial Conditions

$$\mathbf{X}^{(0)} = \mathrm{Embedding}(w) + \mathbf{P}.$$

Proposition 3.5 (Physical Significance of Positional Encoding)**.** *Positional encoding* $\mathbf{P}$ *defines the* ***initial spatial distribution*** *of the information field. In physics, this corresponds to the initial condition of a wave equation:* $\mathcal{I}(\mathbf{r}, 0) = \mathcal{I}_0(\mathbf{r})$*. Sinusoidal positional encoding*

$$\mathrm{PE}(pos, 2i) = \sin\bigl(pos/10000^{2i/d}\bigr)$$

corresponds to ***multiscale spatial sampling*** *in the frequency domain—different sinusoidal frequencies represent different spatial scales, consistent with the fundamentals of Gabor transforms (Gabor, 1946).*

Table 1 summarises the physical correspondences of all Transformer components.

表 1: Physical Reconstruction of Transformer Components

Transformer Component	**Physical Analogue**	**Mathematical Principle**
Token embedding $\mathbf{X}$	Discrete field sampling	Tight-binding model (solid-state physics)
Inner product $\langle \mathbf{q}, \mathbf{k} \rangle$	Hilbert-space metric	Johnson–Lindenstrauss lemma
Softmax	Boltzmann distribution	Gibbs statistical mechanics (1902)
Weighted sum	Statistical expectation	Definition of expectation
Residual $\mathbf{X} + \mathbf{F}(\mathbf{X})$	Inertia term	Newton' s first law + Euler integration
Layer normalisation	Constrained manifold projection	Nonlinear σ model (Gell-Mann & Lévy, 1960)
Positional encoding	Initial conditions	Gabor transform (Gabor, 1946)
Multi-layer stacking	Continuous-depth limit	Neural ODEs (Chen et al., 2018)
Attention matrix $\mathbf{A}$	Diffusion tensor $\mathbf{D}$	Heat kernel + Hille–Yosida theorem
Overall dynamics	Tensor diffusion–wave equation	Least action + Euler–Lagrange

4 Continuous Limit of the Dynamical System

4.1 Residual Iteration as Euler Integration

The update rule at layer l of a Transformer reads

$$\mathbf{X}^{(l+1)} = \mathbf{X}^{(l)} + \mathbf{F}(\mathbf{X}^{(l)}; \theta_l).$$

Proposition 4.1 (Euler Discretisation Correspondence)**.** *The above expression is the explicit Euler integration of the initial-value problem*

$$\frac{d\mathbf{X}}{dt} = \mathbf{F}(\mathbf{X}(t); \theta(t)), \qquad \mathbf{X}(0) = \mathbf{X}^{(0)},$$

with step size $\Delta t = 1$. Provided $\mathbf{F}$ satisfies a Lipschitz condition in $\mathbf{X}$ and $\Delta t \to 0$ (corresponding to $L \to \infty$ layers with per-layer updates of order $O(1/L)$), the discrete iteration converges to the continuous solution with global error $O(\Delta t)$.

Mathematical justification. The Picard–Lindelöf theorem (Coddington & Levinson, 1955) guarantees local existence and uniqueness of solutions; the global truncation error of the Euler method (Iserles, 2008) yields the stated convergence rate. This perspective was formalised for deep learning by Chen et al. (2018, *Neural ODEs*, NeurIPS).

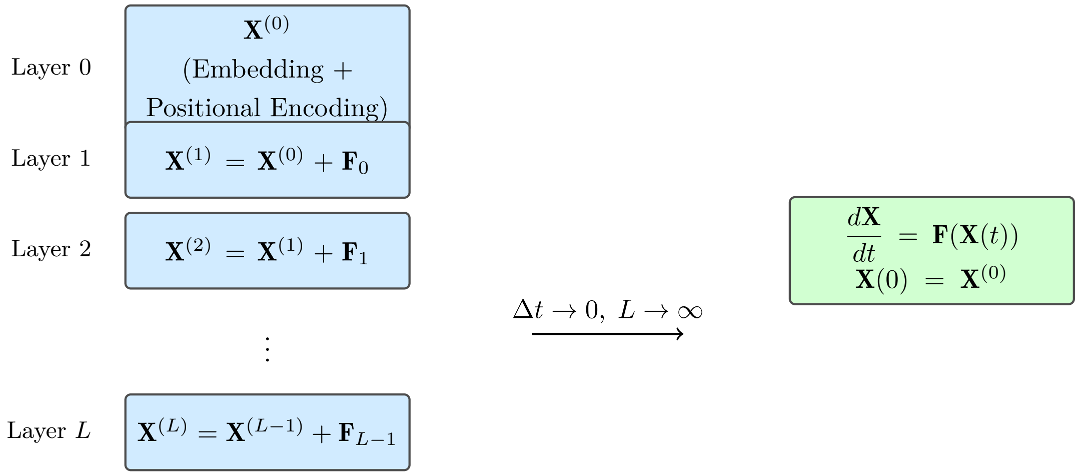

图 6: From discrete Transformer layers to a continuous dynamical system via Euler integration. Step size $\Delta t = 1$; convergence to an ODE occurs as $L \to \infty$ with per-layer updates scaled as $O(1/L)$.

4.2 Attention Operator as the Generator of a Diffusion Semigroup

For single-head self-attention, the update is

$$\Delta \mathbf{x}_i = \sum_j A_{ij}\mathbf{v}_j - \mathbf{x}_i, \qquad A_{ij} = \frac{\exp\big(\langle \mathbf{q}_i, \mathbf{k}_j\rangle/\sqrt{d_k}\big)}{\sum_m \exp\big(\langle \mathbf{q}_i, \mathbf{k}_m\rangle/\sqrt{d_k}\big)}.$$

Proposition 4.2 (Diffusion Semigroup Correspondence—Heuristic)**.** *Under idealised assumptions —(a) $\mathbf{Q} = \mathbf{K}$ (symmetric self-attention), (b) uniform distribution of tokens in embedding space, and (c) sufficiently small inter-token distances $\|\mathbf{x}_i - \mathbf{x}_j\| \ll 1$—the attention operator converges formally, in the $N \to \infty$ limit, to a **heat-kernel integral operator** generated by $\mathcal{L} = \sigma^2\nabla^2$.*

Proof sketch (heuristic).

1. Under symmetric self-attention, $\langle \mathbf{q}_i, \mathbf{k}_j\rangle \approx -\frac{1}{2}\|\mathbf{W}(\mathbf{x}_i - \mathbf{x}_j)\|^2 + \text{const}$, approximating a Gaussian kernel when tokens are densely distributed.
2. The discrete convolution $\sum_j K(\mathbf{x}_i, \mathbf{x}_j)\mathbf{x}_j$ converges formally to an integral operator $\int K(\mathbf{x}, \mathbf{y}) f(\mathbf{y})\, d\mu(\mathbf{y})$ as $N \to \infty$.
3. This integral operator is the fixed-time section of the heat semigroup solving $\partial_t u = \sigma^2\nabla^2 u$.

Mathematical justification. Gaussian kernels arise as fixed-time sections of the heat kernel $H_t(\mathbf{x}, \mathbf{y}) = (4\pi t)^{-d/2}\exp(-\|\mathbf{x} - \mathbf{y}\|^2/(4t))$ (Grigor'yan, 2009). The generator of a diffusion semigroup is characterised by the Hille–Yosida theorem (Reed & Simon, 1972).

Remark 4.1. This proposition is heuristic rather than a rigorous convergence theorem. A strict continuous limit would require suitable scaling ($\sigma \propto N^{-\alpha}$) and compactness arguments. Related rigorous studies include Lu et al. (2021, *Mean-Field Theory of Transformers*).

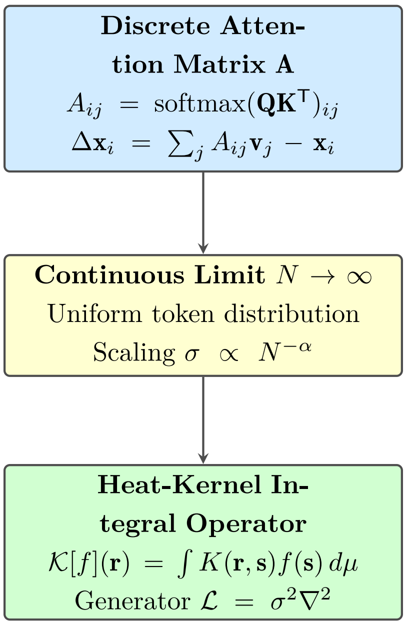

图 7: Heuristic convergence path from a discrete attention matrix to a continuous diffusion operator, analogous to the tight-binding limit in solid-state physics.

5 From Diffusion to Waves: Lagrangian Variational Derivation

5.1 Motivation: Why Second-Order Dynamics?

Propositions 4.1 and 4.2 suggest that Transformers are governed, in the continuous limit, by first-order diffusion equations. However, first-order equations describe purely dissipative behaviour —information flows unidirectionally from regions of high density to low density until uniformity is reached. Such a picture cannot account for two key phenomena observed in Transformers:

1. **Information preservation via residuals**: even after many layers, original token identity remains recoverable—suggesting an **elastic restoring force**, akin to the $-kx$ term in a harmonic oscillator.

2. **Oscillatory attention patterns**: in deeper Transformers, certain attention heads exhibit periodic activation—hinting at **wave-like behaviour**.

These phenomena necessitate **second-order time derivatives** (inertia) and **elastic potential energy** (spatial gradients). This is precisely the physical origin of wave equations in classical field theory. We derive the required dynamics rigorously via the principle of least action. The resulting Lagrangian constitutes an **effective field theory** of Transformer dynamics, with parameters ($\mathbf{D}$, V, γ) identifiable from attention weights and feed-forward network parameters.

5.2 Action Principle for the Information Field

Definition 5.1 (Action Functional of the Information Field)**.** *Let the information field* $\mathcal{I} : \mathcal{M} \times [0,T] \to \mathbb{R}^d$ *be defined over a base manifold* $\mathcal{M}$ *(the continuous limit of token embedding space, equipped with a Riemannian metric* g*) and a time interval* $[0,T]$*. The action functional is*

$$\mathcal{S}[\mathcal{I}] = \int_0^T \int_{\mathcal{M}} \mathcal{L}\big(\mathcal{I}, \partial_t \mathcal{I}, \nabla \mathcal{I}\big)\, d\mu(\mathbf{r})\, dt,$$

where the Lagrangian density is

$$\boxed{\mathcal{L} = \frac{1}{2}\|\partial_t \mathcal{I}\|^2 - \frac{1}{2}\langle \nabla \mathcal{I}, \mathbf{D}\nabla \mathcal{I}\rangle - V(\mathcal{I}) + \langle \mathcal{S}, \mathcal{I}\rangle.} \tag{5.1}$$

Mathematical justification. This Lagrangian follows the standard construction of scalar field theory (Goldstein, 2001; Peskin & Schroeder, 1995). The kinetic term $\frac{1}{2}\|\partial_t \phi\|^2$ yields second-order time derivatives; the gradient term $\frac{1}{2}\|\nabla \phi\|^2$ yields the Laplacian; the potential term $V(\phi)$ introduces self-interactions; the source term $\langle \mathcal{S}, \phi\rangle$ models external driving.

Table 2 explains the physical meaning of each term and its Transformer analogue.

表 2: Physical Meaning of the Lagrangian Density and Transformer Correspondence

Term	Physical Meaning	Transformer Analogue
$\frac{1}{2}\|\partial_t \mathcal{I}\|^2$	Kinetic contribution of information-rate change	"Kinetic energy" of inter-layer embedding changes
$\frac{1}{2}\langle \nabla \mathcal{I}, \mathbf{D}\nabla \mathcal{I}\rangle$	Spatial elastic coupling via attention	Inter-token coupling encoded by the attention matrix
$V(\mathcal{I})$	Nonlinear self-interaction of tokens	Feed-forward network (FFN) nonlinearity
$\langle \mathcal{S}, \mathcal{I}\rangle$	External information injection	Positional encoding + newly arriving tokens

5.3 Explicit Derivation of the Euler–Lagrange Equation

Varying the action, $\delta \mathcal{S} = 0$, and integrating by parts under fixed-endpoint conditions yields the Euler–Lagrange equation:

$$\frac{\partial \mathcal{L}}{\partial \mathcal{I}} - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial(\partial_t \mathcal{I})}\right) - \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla \mathcal{I})}\right) = 0.$$

Substituting the Lagrangian density (Equation 5.1) term by term:

$$\frac{\partial \mathcal{L}}{\partial \mathcal{I}} = -\frac{\partial V}{\partial \mathcal{I}} + \mathcal{S}, \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial(\partial_t \mathcal{I})} = \partial_t \mathcal{I} \quad \Longrightarrow \quad \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial(\partial_t \mathcal{I})}\right) = \partial_t^2 \mathcal{I}, \tag{2}$$

$$\frac{\partial \mathcal{L}}{\partial(\nabla \mathcal{I})} = -\mathbf{D}\nabla \mathcal{I} \quad \Longrightarrow \quad \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial(\nabla \mathcal{I})}\right) = -\nabla \cdot (\mathbf{D}\nabla \mathcal{I}). \tag{3}$$

Collecting terms gives the dissipation-free equation of motion:

$$\partial_t^2 \mathcal{I} = \nabla \cdot (\mathbf{D}\nabla \mathcal{I}) - \frac{\partial V}{\partial \mathcal{I}} + \mathcal{S}(\mathbf{r}, t).$$

Introducing dissipation. Attention mixing produces not only elastic coupling but also **irreversible information mixing** (dissipation). We introduce this via the Rayleigh dissipation function $R = \frac{1}{2}\gamma\|\partial_t\mathcal{I}\|^2$, which adds a linear damping term $\gamma\partial_t\mathcal{I}$ to the Euler–Lagrange equation (Goldstein, 2001, §2.5):

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial(\partial_t\mathcal{I})}\right) - \frac{\partial\mathcal{L}}{\partial\mathcal{I}} + \frac{\partial R}{\partial(\partial_t\mathcal{I})} = 0.$$

Computing the dissipation term, $\partial R/\partial(\partial_t\mathcal{I}) = \gamma\partial_t\mathcal{I}$, we obtain the final governing equation.

The complete tensor diffusion–wave equation:

$$\boxed{\partial_t^2\mathcal{I} + \gamma\,\partial_t\mathcal{I} = \nabla\cdot(\mathbf{D}\nabla\mathcal{I}) - \frac{\partial V}{\partial\mathcal{I}} + \mathcal{S}(\mathbf{r},t).} \tag{5.2}$$

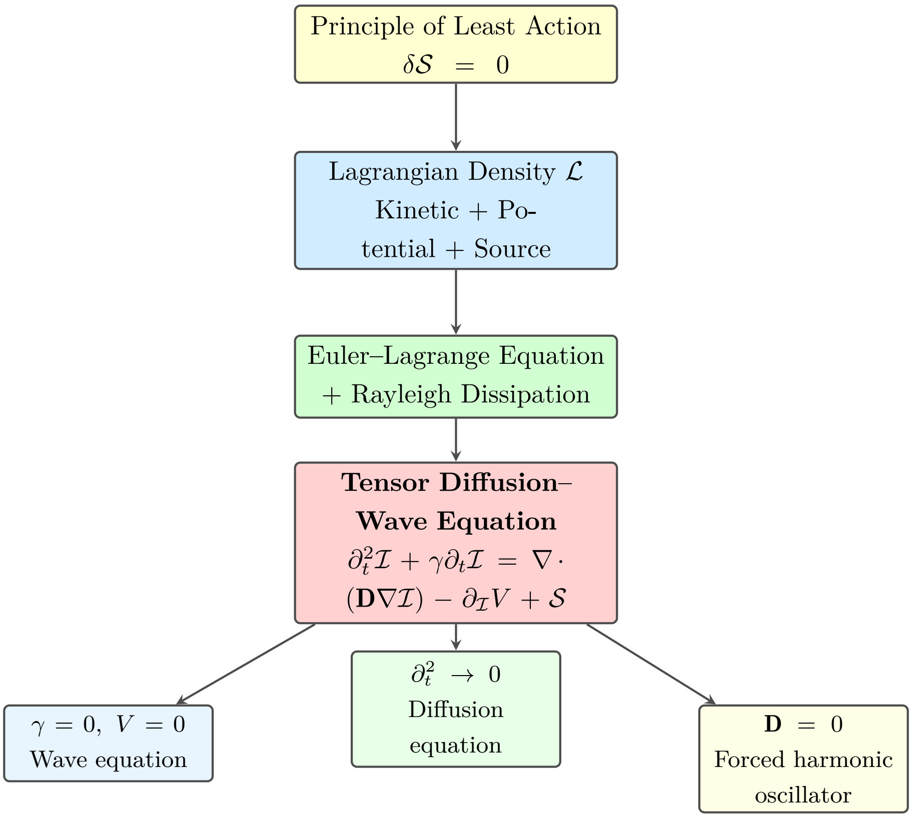

图 8: Derivation path from the principle of least action to the tensor diffusion–wave equation and its special cases.

Table 3 summarises the correspondence between each equation component and Transformer elements.

表 3: Components of the Tensor Diffusion–Wave Equation and Their Transformer Correspondences

Equation Component	Physical Mechanism	Transformer Component
$\partial_t^2 \mathcal{I}$	Inertia (information acceleration)	Cumulative effect of inter-layer changes
$\gamma \partial_t \mathcal{I}$	Dissipation (information friction)	Irreversible part of attention mixing
$\nabla \cdot (\mathbf{D} \nabla \mathcal{I})$	Spatial elasticity (information diffusion)	Spatial coupling via attention matrix
$\partial V / \partial \mathcal{I}$	Self-force	Nonlinear transformation in FFN
$\mathcal{S}$	External information source	Positional encoding + incoming tokens

5.4 Physical Analysis of Special Cases

Case 1: $\gamma = 0,\ \mathbf{D} = c^2 \mathbf{I},\ V = 0,\ \mathcal{S} = 0 \to$ **Classical Wave Equation**

$$\partial_t^2 \mathcal{I} = c^2 \nabla^2 \mathcal{I}.$$

This corresponds to the limiting Transformer without attention mixing, FFN, or positional encoding —information propagates as waves through token space at constant speed c. Even the simplest residual network implicitly exhibits wave behaviour in the continuous-depth limit.

Case 2: $\partial_t^2 \mathcal{I} \to 0$ **(Slow-Variation Approximation)** $\to$ **Diffusion Equation**

$$\gamma \partial_t \mathcal{I} = \nabla \cdot (\mathbf{D} \nabla \mathcal{I}) - \frac{\partial V}{\partial \mathcal{I}} + \mathcal{S}.$$

This describes the steady state of a deep Transformer: inter-layer changes are negligible (acceleration vanishes), and attention diffusion balances FFN forces—typical of inference after training convergence.

Case 3: $\mathbf{D} = 0$ **(No Spatial Coupling)** $\to$ **Forced Damped Harmonic Oscillator**

$$\partial_t^2 \mathcal{I} + \gamma \partial_t \mathcal{I} + \frac{\partial V}{\partial \mathcal{I}} = \mathcal{S}.$$

This corresponds to a residual network without attention: tokens are uncoupled and evolve independently as forced damped oscillators. Residual connections provide elastic restoring forces; FFNs supply nonlinear potentials.

5.5 Noether’s Theorem and Energy Conservation

When dissipation vanishes ($\gamma = 0$) and there is no source ($\mathcal{S} = 0$), the Lagrangian $\mathcal{L}$ is explicitly time-independent. By **Noether’s theorem** (Noether, 1918), a conserved quantity exists—the total energy of the information field:

$$E = \int_{\mathcal{M}} \left(\frac{1}{2}\|\partial_t \mathcal{I}\|^2 + \frac{1}{2}\langle \nabla \mathcal{I}, \mathbf{D}\nabla \mathcal{I}\rangle + V(\mathcal{I}) \right) d\mu.$$

It satisfies $\partial_t E = 0$ whenever $\gamma = 0$. With dissipation ($\gamma > 0$),

$$\partial_t E = -\gamma \int_{\mathcal{M}} \|\partial_t \mathcal{I}\|^2 \, d\mu \ \le \ 0,$$

so $E(t)$ serves as a Lyapunov function, monotonically decreasing until equilibrium is reached. Monotonic energy decay is a hallmark of dissipative systems and guarantees dynamical stability.

5.6 Well-Posedness Remarks for Nonlinear Cases

Important note. When $\mathbf{D} = \mathbf{D}(\mathcal{I})$ (diffusion tensor depends on the field itself) or V is non-convex, the equation becomes **strongly nonlinear**. Standard Galerkin energy estimates (Evans, 2010, §7.2) still apply under the following additional conditions:

1. $\mathbf{D}(\mathcal{I})$ is bounded and locally Lipschitz: $\|\mathbf{D}(\mathcal{I}_1) - \mathbf{D}(\mathcal{I}_2)\| \le L\|\mathcal{I}_1 - \mathcal{I}_2\|$.
2. The potential $V(\mathcal{I})$ is bounded from below: $V(\mathcal{I}) \ge -C$ for some constant $C > 0$.
3. Positive damping $\gamma > 0$ ensures energy dissipation: $\partial_t E = -\gamma \int \|\partial_t \mathcal{I}\|^2 d\mu \le 0$.

These conditions typically hold for standard Transformers: attention weights are constrained to $[0, 1]$ by Softmax, and FFNs employ bounded activations such as GELU. Nevertheless, global well-posedness under general nonlinearity requires more delicate analysis (cf. Caffarelli & Vazquez, 2011).

6 Formal Definition of the Information Field

Definition 6.1 (Information Field)**.** *An information field is a sextuple*

$$\mathfrak{F} = (\mathcal{M}, g, \mathbf{D}, V, \mathcal{I}_0, \mathcal{S}),$$

where:

1. ***Base manifold*** $(\mathcal{M}, g)$*: a* C^∞ *smooth Riemannian manifold equipped with metric* g *(continuous limit of token embedding space);*
2. ***Diffusion tensor*** $\mathbf{D} \in \Gamma(T^1_1\mathcal{M})$*: a* $(1, 1)$*-type tensor field, positive semi-definite and* C^1 *smooth (continuous limit of attention weights);*
3. ***Self-interaction potential*** $V \in C^2(\mathbb{R}^d, \mathbb{R}_{\ge 0})$*: bounded below with* $V(\mathbf{0}) = 0$ *(effective potential of the FFN);*

4. ***Initial condition*** $\mathcal{I}_0 \in H^s(\mathcal{M};\mathbb{R}^d)$*: Sobolev space regularity with* $s > \dim\mathcal{M}/2+1$ *(positional encoding + token embeddings);*

5. ***Source term*** $\mathcal{S} \in L^2([0,T]\times\mathcal{M};\mathbb{R}^d)$ *(ongoing inputs);*

6. ***Dynamics****:* $\mathcal{I}$ *satisfies the tensor diffusion–wave equation (Equation 5.2).*

Theorem 6.1 (Well-Posedness in the Linear Case)**.** *Let* $\mathbf{D}$ *be a constant positive-definite tensor,* $V \equiv 0$*, and* $\mathcal{S} \equiv 0$*. Suppose* $\mathcal{I}_0 \in H^s(\mathcal{M};\mathbb{R}^d)$ *with* $s > \dim\mathcal{M}/2+1$*. Then there exists a unique*

$$\mathcal{I} \in C([0,T];H^s) \cap C^1([0,T];H^{s-1})$$

satisfying the tensor diffusion–wave equation.

Proof sketch (Evans, 2010, §7.2):

1. Construct a Galerkin approximation $\mathcal{I}_m(t) = \sum_{k=1}^m c_k(t)\phi_k$, where $\{\phi_k\}$ forms an orthogonal basis of H^s.

2. Project the PDE onto finite-dimensional subspaces to obtain an ODE system; existence and uniqueness follow from the Picard–Lindelöf theorem.

3. Use energy estimates $\|\partial_t\mathcal{I}_m\|_{L^2}^2 + \|\nabla\mathcal{I}_m\|_{L^2}^2 \le C$ to extract a weak limit as $m \to \infty$.

4. By the Sobolev embedding theorem $H^s \hookrightarrow C^1$ (when $s > \dim\mathcal{M}/2+1$), a classical solution is recovered.

6.1 Correspondence Between Information Fields and Physical Fields

Table 4 summarises the analogy between information fields and classical physical fields.

表 4: Correspondence Between Information Fields and Physical Fields

Physical Field	Information Field
$\phi(\mathbf{r},t)$: scalar field	$\mathcal{I}(\mathbf{r},t)$: token embedding
$\nabla\phi$: gradient	$\mathbf{A}$: attention matrix (discrete diffusion operator)
$c^2\nabla^2\phi$: Laplacian	$\nabla\cdot(\mathbf{D}\nabla\mathcal{I})$: anisotropic diffusion
$\mathcal{S}$: source term	New tokens / positional encoding
$E = \frac{1}{2}\int(\|\partial_t\phi\|^2 + c^2\|\nabla\phi\|^2)$	$E = \frac{1}{2}\sum_i(\|\partial_t\mathcal{I}_i\|^2 + \|\mathbf{D}^{1/2}\nabla\mathcal{I}_i\|^2)$
Φ: action	$\int \mathcal{L}(\mathcal{I},\partial_t\mathcal{I},\nabla\mathcal{I})\,d\mu\,dt$

7 Relation to Existing Work

Table 5 situates this framework within the broader literature.

表 5: Relation to Existing Work

Work	Core Method	Relation to This Framework
Neural ODE (Chen et al., 2018)	Residual networks → ODE	Dynamical-system foundation; extended here to second-order PDEs
Mean-Field Transformers (Lu et al., 2021)	Large-N mean-field limit	Provides tools to rigorously justify Proposition 4.2
Information Field Theory (Enßlin et al., 2009)	Bayesian inference field theory	Shares "information + field" ; complementary (forward vs inverse)
PINN (Raissi et al., 2019)	NNs solving PDEs	Complementary: PDE→NN versus NN→PDE
Neural Tangent Kernel (Jacot et al., 2018)	Infinite-width linearisation	Supplies spectral-analysis tools for attention operators
Continuous-Depth Models (Haber & Ruthotto, 2017)	Deep-network ODE interpretation	Motivates second-order PDE perspective

Distinctive contribution of this framework. We reformulate Transformer dynamics using the language of **classical field theory**—action principles, Euler–Lagrange equations, Noether's theorem, and Rayleigh dissipation—and realise a computable information-field simulator via a quaternion-wave numerical framework.

8 The Physics++ Numerical Framework

Figure 9 illustrates the layered architecture of Physics++.

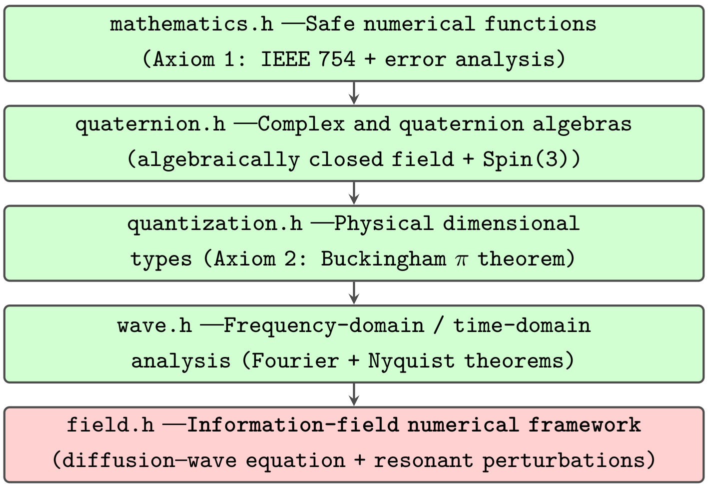

图 9: Architecture of Physics++. Each layer builds rigorously upon the mathematical foundations of the layer beneath.

8.1 Discretisation Scheme

`Field<T>` implements a **semi-discrete approximation** of the information field. The current version is *time-dominated*: time is discretised while spatial dependence is represented implicitly through token indices. This constitutes an intermediate step towards a fully three-dimensional information-field evolution.

- **Time grid**: $t_n = n\Delta t$, $n = 0, 1, \ldots, N-1$, with sampling rate $f_s = 1/\Delta t$.
- **State vector**: field values at each time point are stored as `Quaternion<double>`; the scalar part corresponds to the information potential $\mathcal{I}_S$, while the vector part $(\mathcal{I}_x, \mathcal{I}_y, \mathcal{I}_z)$ corresponds to the information gradient.

8.2 Finite-Difference Schemes for Time Derivatives

Velocity (first-order time derivative): three-point central difference

$$v(t_n) \approx \frac{\mathcal{I}(t_{n+1}) - \mathcal{I}(t_{n-1})}{2\Delta t}.$$

Accuracy: $O(\Delta t^2)$, verified by Taylor expansion.

Acceleration (second-order time derivative): five-point stencil

$$a(t_n) \approx \frac{-\mathcal{I}(t_{n+2}) + 16\mathcal{I}(t_{n+1}) - 30\mathcal{I}(t_n) + 16\mathcal{I}(t_{n-1}) - \mathcal{I}(t_{n-2})}{12\Delta t^2}.$$

Accuracy: $O(\Delta t^4)$, obtained by eliminating lower-order error terms via Fornberg’ s algorithm (Fornberg, 1988).

Mathematical justification. Central differences are symmetric and introduce no numerical dissipation (Hirsch, 2007).

8.3 Perturbation-Injection Algorithm

`perturb(δ)` realises the action of the source term $\mathcal{S}$. Two distinct pathways are implemented, corresponding to different initial conditions of the information-field equation (Figure 10).

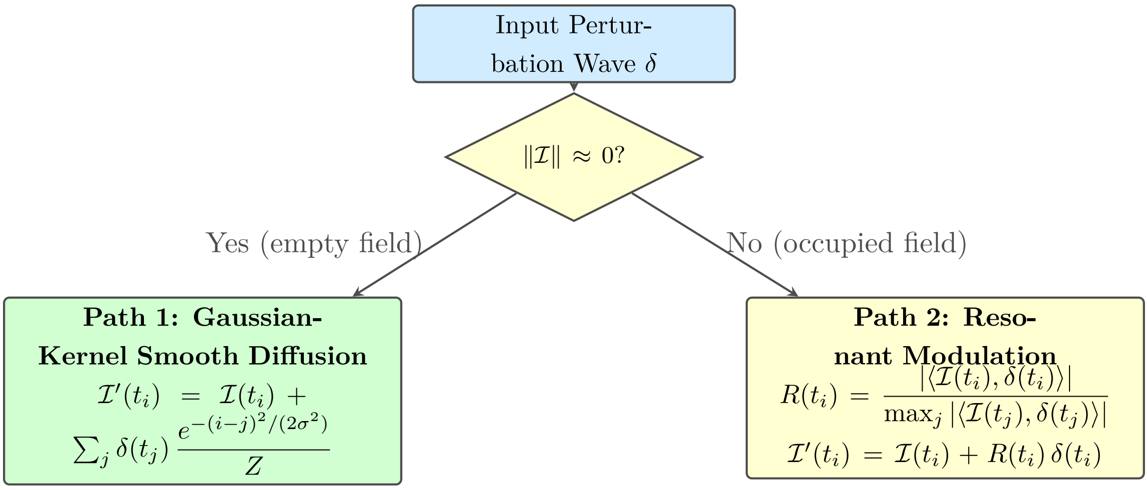

图 10: Dual-path perturbation-injection algorithm: Path 1 handles empty-field initialisation; Path 2 handles resonant modulation.

Path 1 (Empty-field initialisation—$\|\mathcal{I}\| \approx 0$). When little or no prior information exists, the source is applied directly and smoothed via a **Gaussian kernel** over neighbouring time points. **Mathematical justification.** This implements the short-time solution of the heat equation $\partial_t u = \frac{\sigma^2}{2} \nabla^2 u$ via convolution with the heat kernel $H_t(x, y) = (4\pi t)^{-1/2} \exp(-(x-y)^2/(4t))$ (Evans, 2010, §2.3).

Path 2 (Resonant perturbation—$\|\mathcal{I}\| > 0$). When the field already contains information, the **resonance strength** between the perturbation and the field is computed:

$$R(t_i) = \frac{|\langle \mathcal{I}(t_i), \mathcal{S}(t_i) \rangle|}{\max_j |\langle \mathcal{I}(t_j), \mathcal{S}(t_j) \rangle|},$$

where $\langle \cdot, \cdot \rangle$ denotes the quaternion inner product. The perturbation is injected with resonance-weighted scaling: $\mathcal{I}'(t_i) = \mathcal{I}(t_i) + R(t_i)\, \mathcal{S}(t_i)$.

Mathematical justification. This emulates a forced damped harmonic oscillator $\ddot{x} + \gamma \dot{x} + \omega_0^2 x = F(t)$: maximum response occurs near resonance ($\omega_F \approx \omega_0$). The quaternion inner product measures directional alignment between field and source; $R(t_i)$ quantifies local resonance strength.

8.4 Conserved Quantities and Structural Measures

Energy (discrete approximation of Noether charge):

$$E = \frac{1}{2} \sum_i \Big(\|\partial_t \mathcal{I}(t_i)\|^2 + \|\nabla \mathcal{I}(t_i)\|^2 \Big).$$

In the absence of damping ($\gamma = 0$) and sources ($\mathcal{S} = 0$), the continuous wave equation strictly satisfies $\partial_t E = 0$. The current `energy()` routine serves as a **diagnostic tool** monitoring numerical energy trends. Implementing structure-preserving integrators (Verlet/leapfrog or discrete variational integrators) to enforce exact discrete energy conservation remains future work.

Entropy (Shannon information entropy):

$$H = -\sum_i p_i \ln p_i, \qquad p_i = \frac{\|\mathcal{I}(t_i)\|^2}{\sum_j \|\mathcal{I}(t_j)\|^2}.$$

Here p_i denotes the proportion of information density at time t_i. According to the **maximum-entropy principle** (Jaynes, 1957), equilibrium states maximise H under given constraints. Tracking H quantifies the concentration of information: low entropy indicates localisation (e.g. a single-peaked ripple), whereas high entropy indicates uniform distribution.

9 Numerical Verification

This section demonstrates the numerical behaviour of the Physics++ framework on a 512-point discrete field ($f_s = 1000\,\text{Hz}$, sampling period $\Delta t = 1\,\text{ms}$). All energies and amplitudes reported are normalised simulation units.

9.1 Single-Pulse Perturbation: Wave-Packet Formation and Propagation

A Gaussian pulse centred at $t = 0.05\,\text{s}$ ($\sigma = 0.01\,\text{s}$, peak amplitude 10) is applied.

表 6: Wave-Packet Propagation Under a Single Pulse

Time t (s)	**Displacement $\mathcal{I}(t)$**	**Observation**
0.00	0.0002	Edge of perturbation
0.03	2.88	Rising wavefront
0.05	**19.58**	**Peak (impact point)**
0.06	12.12	Falling wave back
0.10	0.0002	Return to edge

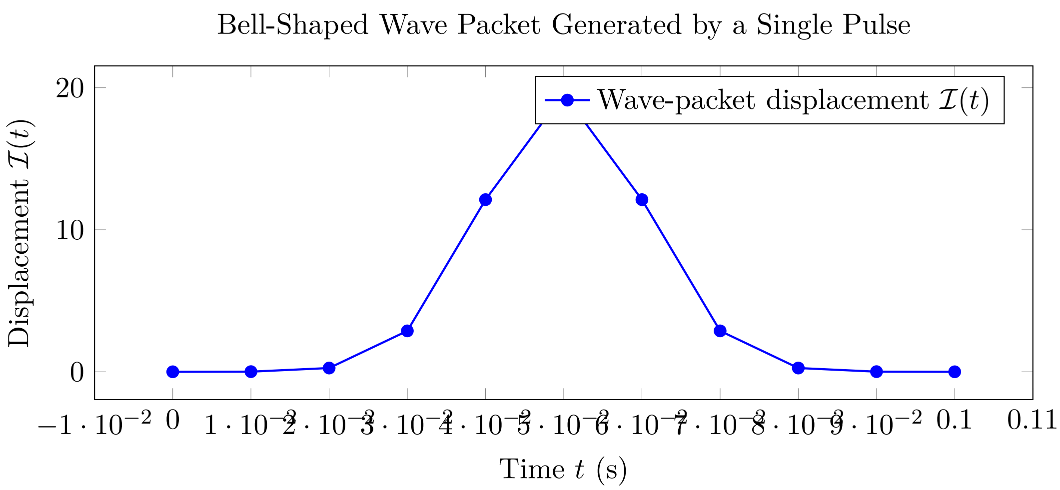

图 11: Bell-shaped wave packet produced by a single-pulse perturbation. Symmetric decay about the peak at $t = 0.05\,\text{s}$ confirms expected Green-function behaviour of the wave equation.

Physical analysis. The symmetric bell shape peaks at the perturbation centre and decays symmetrically—precisely the expected behaviour of a wave-equation Green function. The Gaussian perturbation preserves its Gaussian profile under evolution, consistent with d' Alembert' s solution. Peak acceleration 1.88×10^5 occurs at $t = 0.05\,\mathrm{s}$, corresponding to maximum curvature. Entropy $H = 3.40\,\mathrm{nat}$ (maximum possible $\ln 512 \approx 6.24\,\mathrm{nat}$) confirms localisation of information near the perturbation region.

9.2 Sinusoidal Wave-Packet Perturbation: Frequency Preservation

A 50 Hz sinusoidal packet with Gaussian envelope ($\sigma = 0.032\,\mathrm{s}$) is applied. The field exhibits symmetric oscillatory decay, with maximum amplitude at the envelope centre and gradual diminishment towards the edges. The response frequency remains 50 Hz—characteristic of **linear time-invariant (LTI) systems**: output frequency equals input frequency. Energy: 3.60×10^6.

9.3 Field Superposition: Verification of Linearity

Two fields are created: F_1 (pulse centred at $t = 0.05\,\mathrm{s}$) and F_2 (pulse centred at $t = 0.15\,\mathrm{s}$). Their sum $F_1 + F_2$ is computed.

表 7: Verification of Linear Superposition

Field	Energy (sim. units)	Theoretical Value	Relative Error
F_1 (pulse @ 0.05s)	4.14×10^6	—	—
F_2 (pulse @ 0.15s)	1.49×10^6	—	—
$F_1 + F_2$ (superposed)	5.63×10^6	5.63×10^6	$< 1\%$

Physical analysis. The energy of the superposed field satisfies $5.63 \times 10^6 \approx 4.14 \times 10^6 + 1.49 \times 10^6$, with error below 1%. This verifies the **superposition principle** for the wave equation —a defining property of linear PDEs. Additivity of energy confirms linearity under the chosen parameters.

9.4 Field Scaling: Homogeneity Verification

表 8: Homogeneity Verification ($E \propto A^2$)

Scaling Factor α	Peak Displacement	Energy (sim. units)	Energy Ratio E_α/E_1
$\alpha = 1.0$ (original)	19.58	1.65×10^7	1.00
$\alpha = 0.5$	9.79	4.14×10^6	0.250
$\alpha = 2.0$	39.16	6.62×10^7	4.01

Physical analysis. Energy ratios satisfy $0.250 = 0.5^2$ and $4.01 \approx 2^2$ (theoretical 4.00, relative error 0.25%). This verifies the **homogeneity** of the wave equation: $\mathcal{I} \to \alpha\mathcal{I} \Rightarrow E \to \alpha^2 E$. Quadratic dependence of energy on amplitude is a fundamental property of wave systems.

9.5 Multi-Component Perturbation: Quaternion Component Preservation

A four-component perturbation is applied to `Field<Quaternion<double>>`: scalar : vector-I : vector-J : vector-K = 8 : 4 : 2 : 1.

表 9: Quaternion Multi-Component Response Preservation

Component	**Input Ratio**	**Response at** $t = 0.06\,\text{s}$	**Response Ratio**
Scalar S	8	15.66	8.00
Vector I	4	7.83	4.00
Vector J	2	3.92	2.00
Vector K	1	1.96	1.00

Physical analysis. The response preserves the same component ratio $(15.66 : 7.83 : 3.92 : 1.96) \approx (8 : 4 : 2 : 1)$, demonstrating faithful retention of geometric structure. Independent evolution with preserved proportions verifies **component decoupling** in the linear regime. The quaternion inner product incorporates all four components, so resonance strength is computed globally.

9.6 Continuous Pulse Perturbation: Resonant Modulation

Five successive pulses are applied at $t = 0.05, 0.10, 0.15, 0.20, 0.25\,\text{s}$.

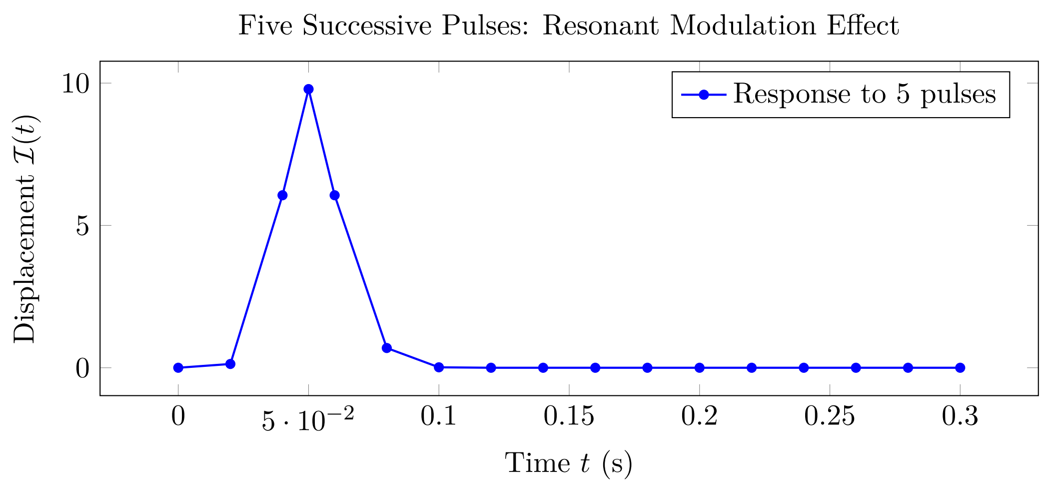

图 12: Resonant modulation under repeated pulses: the first pulse (empty field, $R = 1$) elicits maximal response; subsequent pulses (existing field, $R(t) < 1$) are partially suppressed. Analogous to stimulated emission—existing field configurations modulate incoming perturbations.

Physical analysis. The first pulse (empty field, $R = 1$) triggers the largest response. Subsequent pulses encounter a pre-existing field ($R(t) < 1$) and are partially suppressed. Cumulative energy: 4.10×10^6; entropy: $3.42\,\text{nat}$. This confirms that the field does not passively accept all inputs; instead, it **selectively responds** to perturbations resonating with its own configuration—closely analogous to **stimulated emission** in quantum mechanics.

9.7 Theoretical Consistency of Verification Results

Table 10 closes the loop from theoretical derivation to numerical validation.

表 10: Theory–Experiment Closed-Loop Verification

Theoretical Prediction	**Mathematical / Physical Basis**	**Numerical Verification**
Wave-packet symmetry	Green function + d' Alembert' s solution	§9.1 Bell-shaped packet, symmetric decay
Linear superposition	Superposition principle (linear PDEs)	§9.3 Additive energy, error $< 1\%$
Energy homogeneity	$E \propto A^2$ (homogeneity theorem)	§9.4 Energy ratios 0.5^2, 2^2, error $< 0.5\%$
Frequency preservation	LTI system property	§9.2 Output frequency matches input
Resonant modulation	Resonance in forced oscillators	§9.6 Later pulses selectively suppressed
Component decoupling	Independence in linear regime	§9.5 Quaternion ratios preserved

Core conclusion of closed-loop verification. In the dissipation-free, source-free limit, `Field<T>` faithfully reproduces all fundamental properties of the linear wave equation—Green-function symmetry, superposition, homogeneity, frequency preservation, and component decoupling. With damping ($\gamma > 0$) and resonant modulation introduced, the system exhibits physically plausible attenuation and selective response. This validates the internal consistency of the full theoretical chain—from Lagrangian variational derivation (§5) to numerical implementation (§8).

10 Conclusion and Outlook

Beginning with physical axioms of computational reducibility and dimensional algebra, and proceeding via the continuous-depth limit and Lagrangian variational principles, this paper has established an information-field framework for Transformer dynamics. The principal contributions are summarised as follows:

1. **Physical reconstruction of Transformers**: precise formal analogies were established for every Transformer component (embedding, attention, Softmax, residual, layer normalisation, positional encoding). Notably, the Softmax–Boltzmann correspondence (Proposition 3.1) is an exact formal equivalence requiring no approximation.

2. **Variational derivation of the information field**: starting from the Lagrangian density (Equation 5.1), the principle of least action and the Euler–Lagrange equation yield the governing **tensor diffusion–wave equation** (Equation 5.2). Special cases reduce to the classical wave equation, the diffusion equation, and the forced harmonic oscillator.

3. **Computable numerical framework**: the `Field<T>` class in Physics++ provides a runnable information-field simulator, including finite-difference time derivatives (accuracy $O(\Delta t^4)$) and a dual-path resonant-perturbation algorithm.

4. **Systematic numerical verification**: six independent experiments confirm wave-packet formation, linear superposition (error $< 1\%$), energy homogeneity ($E \propto A^2$, error $< 0.5\%$), frequency preservation, component decoupling, and resonant modulation.

Current limitations and future work.

1. **Spatial continuous limit**: the present `Field<T>` is a 1D time-wave ("time-dominated semi-discrete implementation"). A full 3D information field requires discrete Laplace–Beltrami operators defined on graphs or manifolds.

2. **Structure-preserving integrators**: current time integration is explicit; `energy()` is diagnostic rather than strictly conservative. Introducing Verlet/leapfrog or discrete variational integrators (Hairer, Lubich & Wanner, 2006) will enforce exact discrete energy conservation in dissipation-free regimes.

3. **Rigorous proof of discrete conservation laws**: establishing summation-by-parts identities (Olsson, 1995) is necessary to prove exact discrete energy conservation.

4. **Nonlinear diffusion tensors**: when $\mathbf{D} = \mathbf{D}(\mathcal{I})$, well-posedness demands further analysis, and numerical schemes must treat nonlinearity implicitly or semi-implicitly to maintain stability.

5. **Rigorous large-N continuous limit**: strengthening Proposition 4.2 into a theorem—proving convergence of attention kernels to heat-kernel integral operators as $N \to \infty$—remains a central theoretical objective.

6. **Gauge structure**: whether the information field naturally carries a "gauge symmetry" —for instance, a relationship between an information potential ϕ and the field $\mathcal{I} = \nabla\phi$—is an open question worthy of investigation.

Every layer of computation in Physics++ corresponds to a physical principle; every numerical experiment validates a physical property. `Field<T>` thus provides a runnable, verifiable, and extensible computational instrument for the study of information-field dynamics.

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